iLab 14-01 · Industry partner: Decidr · 36105 iLab Capstone

Assessing the Credibility of Organisational Knowledge from Fragmented Data

An explainable retrieval, stance and Bayesian scoring pipeline for claims about an organisation

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PRESENTATION OUTLINE

Agenda

1. **Research problem and motivation**
   Fragmented, contradictory company knowledge; research question and objectives
2. **Methodology and proposed approach**
   Data and ingestion, architecture, hybrid retrieval, LLM stance, weighting, Bayesian scoring, guardrails, evaluation
3. **Key findings and outcomes**
   Ablation results, error analysis, worked example
4. **Challenges and how they were addressed**
   Technical problems and fixes; threats to validity
5. **Conclusions and project impact**
   What was achieved, impact for Decidr, future work

Research question, motivation and constraints

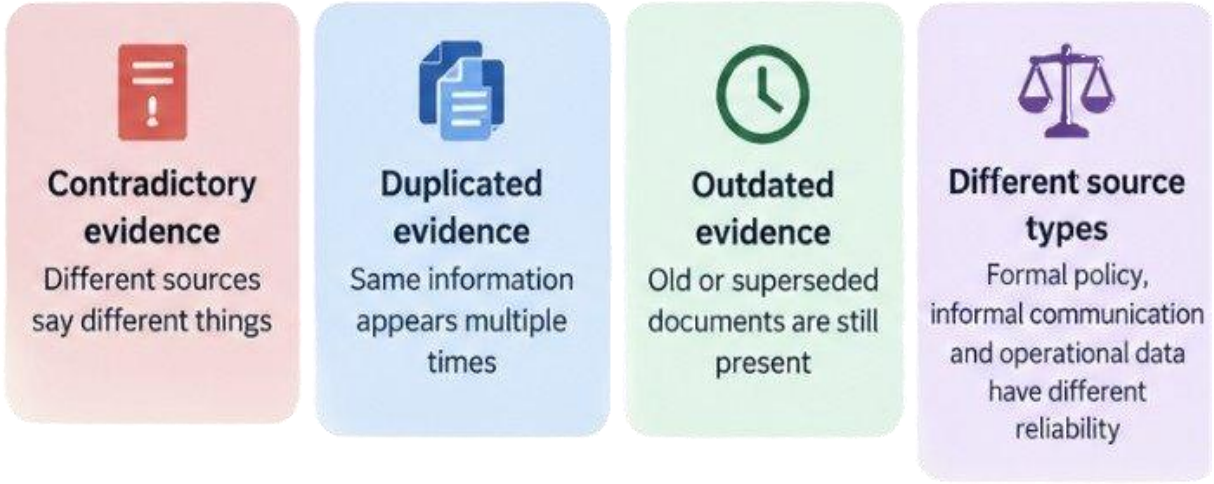
RESEARCH QUESTION

How can the credibility of an organizational knowledge claim be assessed when the available evidence is fragmented, incomplete, contradictory, duplicated, outdated, and comes from different source types?

Dataset: Decidr's Northfield Software Ltd, Phase 1 + Phase 2 data packs

Motivation

Existing search systems can retrieve relevant information, but they usually return a single answer and hide the disagreement between sources. They do not reliably account for:



What we mean by credibility

The brief from Decidr

1. **What they shared:** Northfield Software Ltd: a realistic, made-up company delivered in two data packs.
2. **What's in it:** 45 documents: policies, process docs, interviews, meetings, Slack and email, CRM deals, tasks and an audit log, plus labelled example claims.
3. **The original goal:** A transparent way to judge what an organisation can defensibly claim about how it works: a score with evidence, not a chatbot.

WHAT IS CREDIBILITY?

How strongly a company's own records justify believing a claim today.

It is not the same as:

Relevance a file can mention the topic without supporting the claim

Popularity ten copies of one Slack message are still one source

Formality alone a policy can say one thing while practice does another

Four questions we ask of every piece of evidence

Who said it?

Authority: approved policy vs chat message

When?

Recency: current, or replaced by something newer

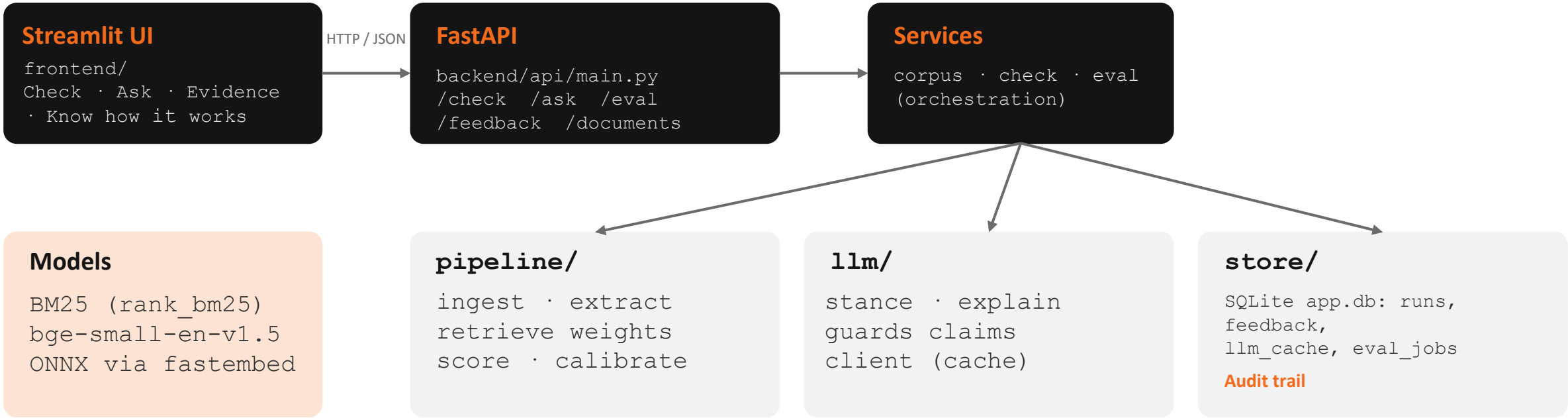
Is it independent?

Copies of the same message count once

Does it agree?

Supports, contradicts, or doesn't address the claim

System architecture



Design decisions

- UI and evaluation call the same engine, so the demo is exactly what is measured
- LLM responses cached by SHA-256(task + prompt): reruns are reproducible and free
- Runs on a CPU laptop; Docker Compose and a run script for hand-over

Data and ingestion

45
source documents

743
chunks
(citable passages)

29
people in roster

Ingestion rules (backend/pipeline/ingest.py)

- **Removed duplicates:** Phase 2 repeats every Phase 1 file, so we keep only one copy of each document (the newer one).
- **Cut files into small, citable pieces:** text documents are split at each heading; spreadsheets (CRM deals, tasks, audit log) are split one row at a time.
- **Separated the email and Slack log:** communications_log.md is split into individual conversation threads (COMMS-01, COMMS-02, etc.)
- **Labelled every piece:** each one records its date, whether it is a draft or out of date, and an ID we can cite (e.g. DEAL-014).

Chunk schema

```
Chunk(
  chunk_id    = "PROC-02::1",
  doc_id      = "PROC-02",
  cite_id     = "PROC-02",
  source_type= "process_doc",
  phase, heading, text,
  metadata    = {superseded, draft, date})
```

Eight-stage pipeline

1	2	3	4	5	6	7	8
Ingest	**Extract**	**Retrieve**	**Reason**	**Weight**	**Score**	**Calibrate**	**Explain**
`ingest.py`	`extract.py`, `llm/claims.py`	`retrieve.py`	`llm/stance.py`	`weight.py`	`score.py`	`calibrate.py`	`llm/explain.py`, `guards.py`
chunks + flags	claims; question → claim	top-15 (RRF)	stance, conf., quote	final weight w_i	p, 90% CI, status	Brier, ECE (n ≥ 30)	checked explanation

LLM

Separation of concerns: the LLM reads, Python scores

- The LLM is used only for language tasks (orange): reason labelling (stage 4) and explanation text (stage 8).
- Weights, the score, the interval and the status are deterministic functions of the labelled evidence: same input, same verdict.
- Every stage has a keyword/template fallback, so the system still runs without an API key.

Retrieval: hybrid lexical + dense search with RRF

Lexical: BM25

- rank_bm25 over tokenised chunks
- Stopword removal, stemming
- Strong on exact terms: "15%", "POL-01", role names

Dense: embeddings

- BAAI/bge-small-en-v1.5 (384-d), ONNX via fastembed
- Cosine similarity, embeddings cached to .npy
- Strong on paraphrase: "sign-off" ≈ "approval"

Reciprocal Rank Fusion

$$\mathrm{RRF}(d) = \sum_{m \in \{\mathrm{bm25},\,\mathrm{dense}\}} \frac{1}{k + \mathrm{rank}_m(d)}$$

$$k = 60, \quad \mathrm{top_k} = 15$$

Rank-based, so BM25 scores and cosine similarities need no calibration against each other.

Measured recall@15 (share of human-expected source documents retrieved, 25 gold claims)

0.79 BM25 only

0.78 Hybrid (RRF)

0.96* Earlier BM25 + rulebook must-fetch

Reason classification: LLM or keyword baseline

Baseline: keyword

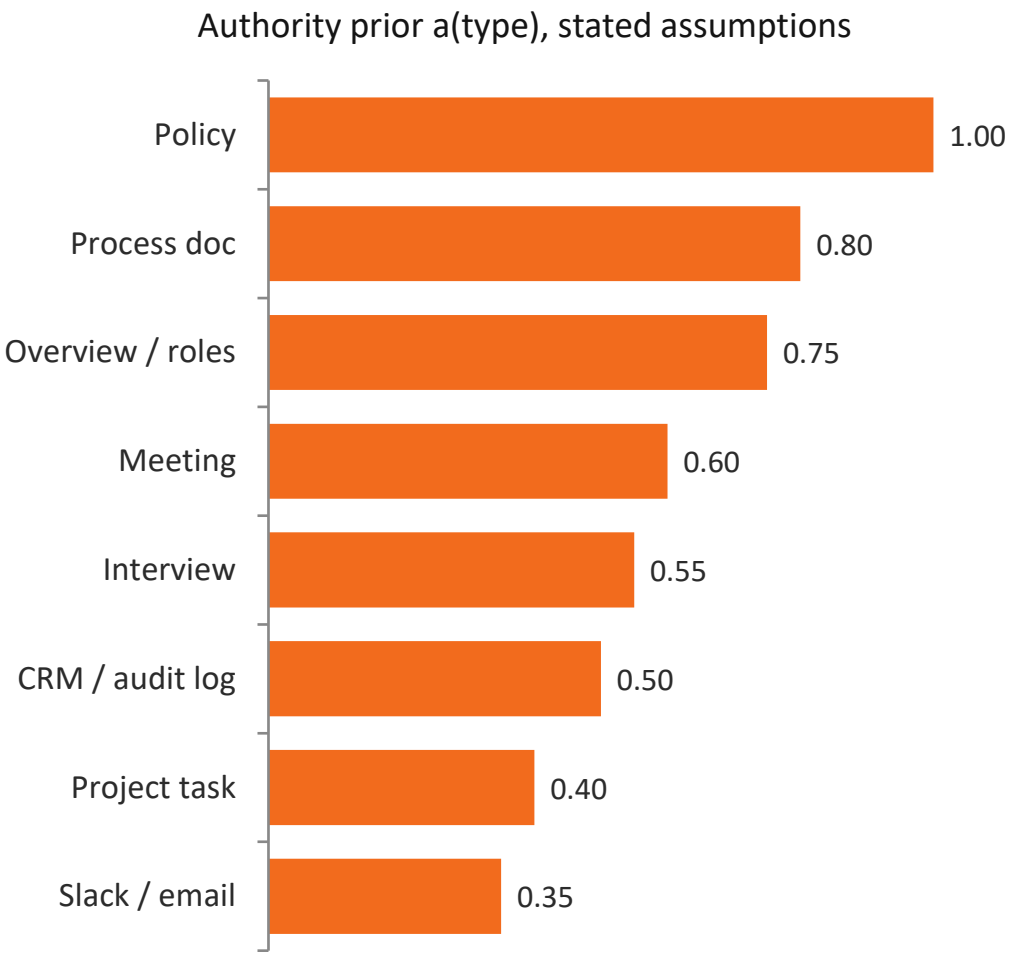
- Lexical overlap between claim and passage
- Negation cues within a small window ("not", "without", "no longer")
- Topic-coherence check and overlap threshold
- Fully deterministic, no API needed

LLM (llm/stance.py)

- One batched request per claim: 15 excerpts (≤ 1,000 chars each) with source type, date, superseded/draft flags
- Rules in prompt: judge the claim as worded; distinguish FORMAL rules from ACTUAL PRACTICE; gap claims; never invent facts
- Temperature 0.1; Gemini with LLM fallback; responses cached by SHA-256 of the prompt

```
{"judgements": [{"ref": "E1",
  "stance": "support|contradict|unrelated",
  "confidence": 0.0-1.0,
  "rationale": "...", "quote": "exact words"}]}
```

Evidence weighting



$$w = a(type) \cdot s \cdot r(t) \cdot (1/n) \cdot c$$

authority × status penalty × recency × independence × judge confidence

s_i — **Status penalty**

0.3 if superseded, 0.5 if draft, else 1

r(t) — **Recency**

1.0 within 365 days of the assessment date, then −0.1 per year, floor 0.7

1/n — **Independence**

Near-duplicates (difflib ratio ≥ 0.82) form a cluster of size n; copies share one vote

c_i — **Confidence**

Stance confidence returned by the judge, in [0, 1]

Bayesian scoring and status assignment

Log-odds update from a neutral prior

$$\text{logit}(p) = \text{logit}(\pi_0) + \Sigma\, w_i^{+} - \Sigma\, w_j^{-}$$
$$p = 1 / (1 + e^{(-\text{logit}(p))}), \quad \pi_0 = 0.5$$

90% interval: bootstrap, 400 resamples of the evidence set (5th–95th percentile)

Status rules (applied in order)

#	Condition	Status
1	$\Sigma w < 0.3$	**Not enough evidence**
2	$\min(\Sigma w^{+}, \Sigma w^{-}) / \Sigma w \geq 0.35$	**Conflicted**
3	$p \geq 0.75$ / $p \geq 0.60$	**Supported / Likely**
4	$p \leq 0.25$ / $p \leq 0.40$	**Unsupported / Unlikely**
5	otherwise	**Conflicted**

Score → status scale



Confidence (separate from status)

- High: Σw ≥ 1.2, ≥ 3 independent clusters, ≥ 2 source types
- Medium: Σw ≥ 0.3
- Low: otherwise

Calibration

Explanation guard (llm/guards.py)

The LLM writes {headline, summary, why[], what_would_change_it} from a fact sheet. check_explanation() rejects it if:

- A required field is missing
- It names a status other than the computed one
- It quotes a number not in the fact sheet
- It cites a source id that was not in the evidence

On rejection: deterministic template explanation, with the reason shown to the user.

Calibration gate (pipeline/calibrate.py)

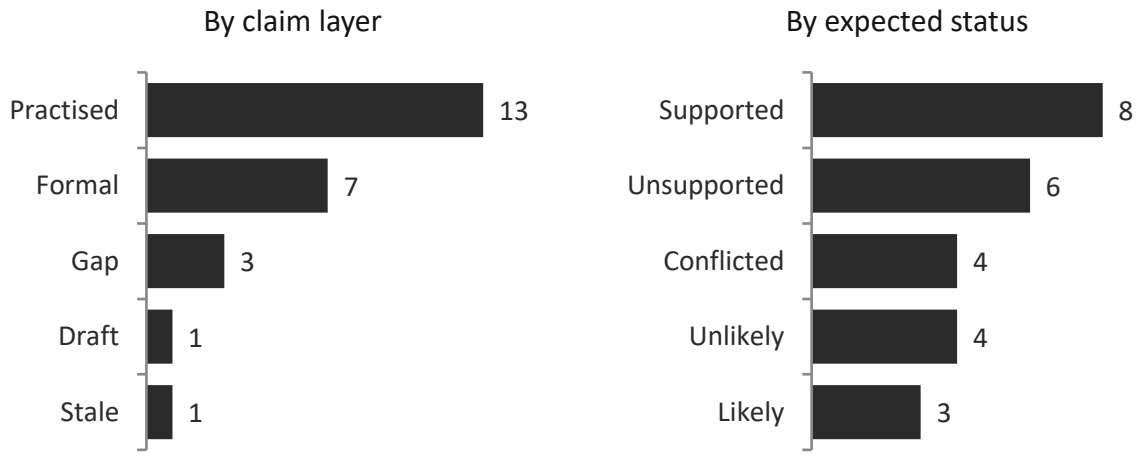
$$\text{Brier} = \frac{1}{N} \sum (p_i - y_i)^2$$

$$\text{ECE} = \sum_b \frac{|B_b|}{N} \cdot |\text{acc}_b - \text{conf}_b|$$

- Labels: true/false outcomes from team gold claims plus in-app user feedback
- Refuses to report Brier, ECE or a reliability table below 30 labels: a curve on fewer points is noise, not evidence
- Current state: below threshold, so calibration is reported as pending

Evaluation protocol

Gold set: hand-labelled claims



Each claim records the expected status, expected supporting and contradicting doc ids, and its business process (P1–P8).

Metrics

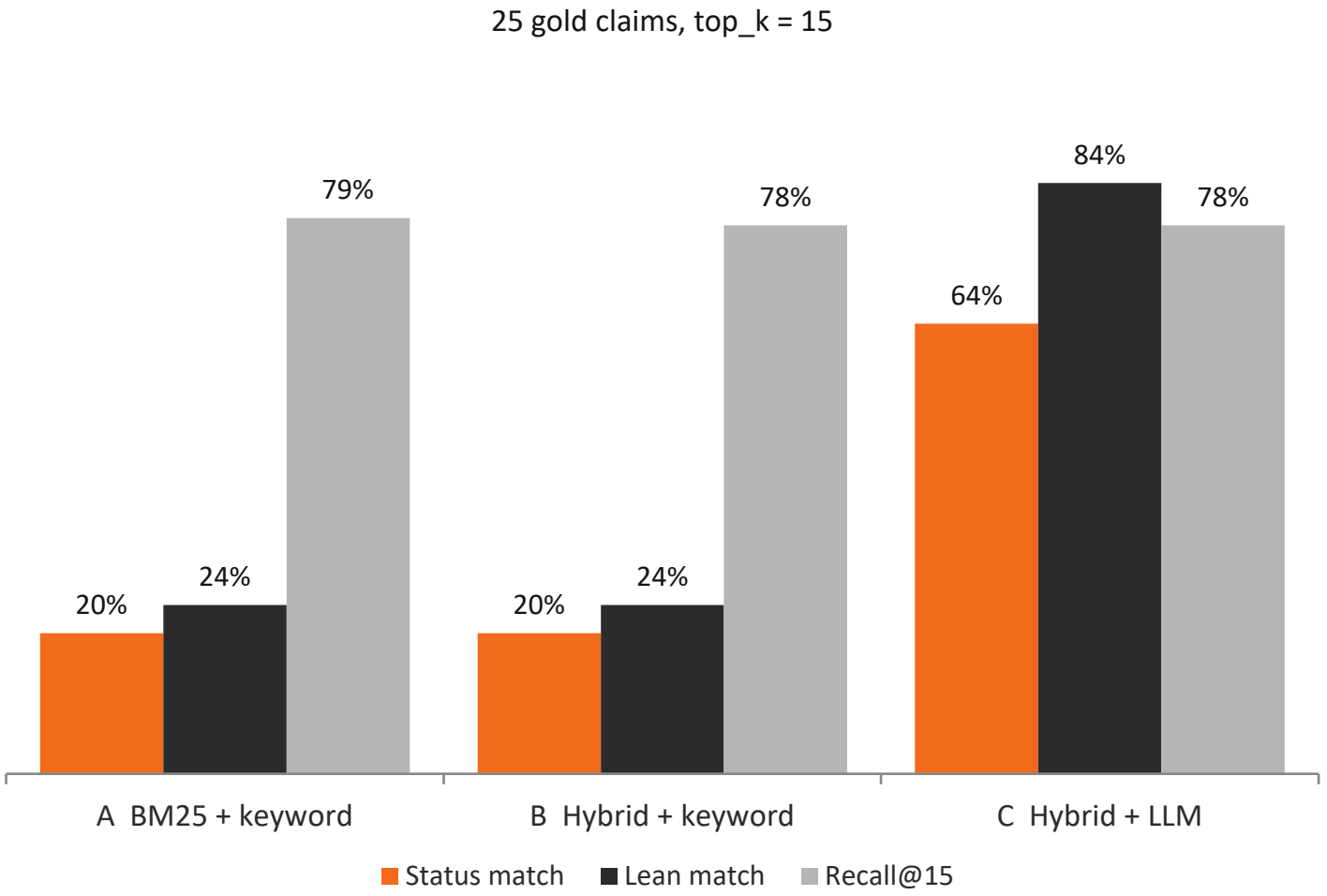
`recall@15`	share of expected doc ids in the top-15 retrieved passages
`status match`	exact agreement with the expected status (6 classes)
`lean match`	agreement on direction: true-ish {Sup, Lik}, false-ish {Unl, Uns}, conflicted, none

Ablation: three configurations, one variable at a time

- A BM25 + keyword judge (baseline)
- B Hybrid RRF + keyword judge (tests retrieval)
- C Hybrid RRF + LLM judge (tests stance)

Fixed: top_k = 15, prior 0.5, same weights and thresholds. Run via POST /eval.

Results: stance quality is the bottleneck, not retrieval



3.2× exact status match, A → C (20% → 64%)

84% verdicts point the right way with the LLM judge

±0 status change from hybrid retrieval alone (A → B)

Worked example: end-to-end trace

c = “Discounts above 15% require Finance approval before a deal is closed.”

1	**Retrieve**	Hybrid RRF, 15 passages. Top hit PROC-02::1 (BM25 rank 2, dense rank 1, cosine 0.83)
2	**Reason**	(1 batched call): 10 support, 2 contradict, 3 unrelated
3	**Weight**	$\Sigma w^{+} = 5.66$, $\Sigma w^{-} = 0.61$ (POL-02, COMMS-05 cite the old 10% rule)
4	**Score**	`logit(p) = 0 + 5.66 − 0.61 = 5.05 → p = 0.994`
5	**Status**	`minority 0.61 / 6.27 = 0.10 < 0.35; p ≥ 0.75 → Supported`
6	**Explain**	LLM explanation passed the guard; cites POL-01, MTG-01, PROC-02

VERDICT

Supported

0.99

CI 0.95 – 1.00
Confidence: High
Evidence: 10 for, 2 against

Superseded 10% sources are shown, not hidden.

Technical challenges and how we addressed them

Stance misclassification

Keyword judge read "without" as negation; 14 of 20 Week-9 errors were false Conflicted.

✓ **LLM with strict JSON, verbatim-quote validation; keyword judge kept as baseline.**

Stale vs current documents

PROC-02 (2022) scored as highly as POL-01 (2025) in retrieval.

✓ **superseded / draft flags at ingest; ×0.3 / ×0.5 penalties; superseded flag passed to the judge.**

Off-topic queries

Stop-words dominated BM25, giving a cited High-confidence verdict on an unrelated question.

✓ **Stopword removal and stemming; Not enough evidence when Σw < 0.3.**

Questions, not claims

"Can I book a $700 flight?" has no lexical overlap with the expense policy.

✓ **Question -> claim rewriting (/ask), with the interpreted claim shown to the user.**

Repetition as corroboration

Copied Slack messages counted as independent sources.

✓ **Near-duplicate clustering (ratio ≥ 0.82); independence weight 1/n.**

Free-tier API limits

Gemini free tier caps requests per day.

✓ **One batched request per claim, SHA-256 response cache, model fallback chain.**

Conclusions and project impact

1

Answered the research question

An explainable pipeline reaches 64% exact and 84% directional agreement with human labels; the LLM judge tripled exact agreement over the keyword baseline.

2

Explainable by construction

The LLM only labels and writes; scores, intervals and statuses are deterministic, guarded and reproducible from the SQLite audit trail.

3

Conflicts and gaps are findings

Conflicted and Not enough evidence expose where policy and practice diverge, which is often what an organisation most needs to know.

Impact for Decidr

- Data-driven pipeline: another organisation's files can be ingested without code changes
- Every verdict traceable to files a non-expert can open

Future work

- Phase 3 held-out test
- Local LLM option (e.g. Ollama) for data privacy
- Fit weights and report calibration at n ≥ 30

Thank you. Questions?

iLab 14-01 · Northfield Credibility · with thanks to Decidr and our mentor

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